Feature Selection for Numerai Classic Tournament

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Numerai NYC Community Meetup
2022-09-24

Main Model - IA_AI

"The NMR Life Chose Me."

My NFTees


New Models

Background

Civil Engineering

Work

Data Scientist / Meetup Organizer

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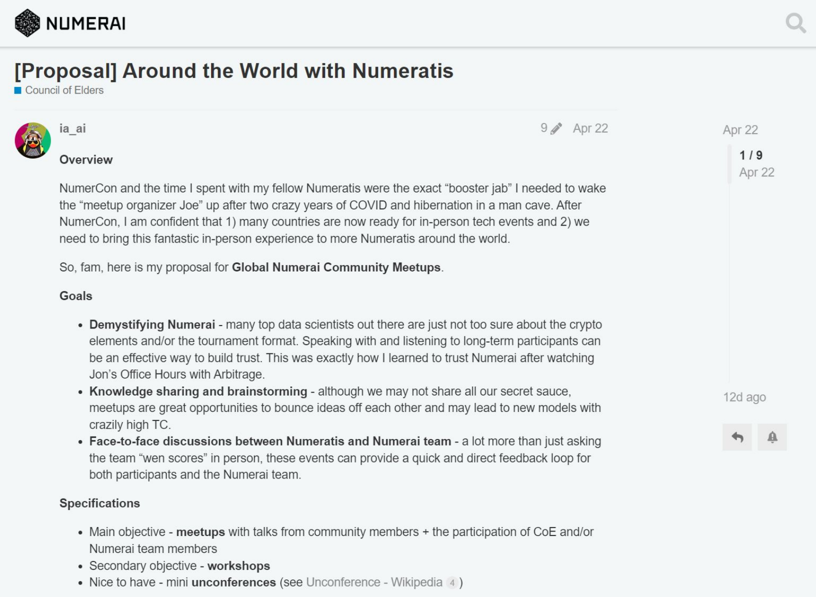
Numerai Community Meetups

How It Started

NumerCon 2022 (April 2022)



NUMERAI

[Proposal] Around the World with Numeratis

Council of Elders

ia_ai 9 Apr 22

Overview

NumerCon and the time I spent with my fellow Numeratis were the exact "booster jab" I needed to wake the "meetup organizer Joe" up after two crazy years of COVID and hibernation in a man cave. After NumerCon, I am confident that 1) many countries are now ready for in-person tech events and 2) we need to bring this fantastic in-person experience to more Numeratis around the world.

So, fam, here is my proposal for **Global Numerai Community Meetups**.

Goals

- **Demystifying Numerai** - many top data scientists out there are just not too sure about the crypto elements and/or the tournament format. Speaking with and listening to long-term participants can be an effective way to build trust. This was exactly how I learned to trust Numerai after watching Jon's Office Hours with Arbitrage.
- **Knowledge sharing and brainstorming** - although we may not share all our secret sauce, meetups are great opportunities to bounce ideas off each other and may lead to new models with crazily high TC.
- **Face-to-face discussions between Numeratis and Numerai team** - a lot more than just asking the team "wen scores" in person, these events can provide a quick and direct feedback loop for both participants and the Numerai team.

Specifications

- Main objective - **meetups** with talks from community members + the participation of CoE and/or Numerai team members
- Secondary objective - **workshops**
- Nice to have - mini **unconferences** (see Unconference - Wikipedia 4)

Apr 22

1 / 9

Apr 22

12d ago

Worldwide Meetups Proposal

How It’s Going

(Reminder: Take a 360 pic)

London (July 2022)

NYC (September 2022)

Numerai Tournament
(you already heard about it from Alex)

V4 Data
(and how I slice it)

V4 Data (Link 1 & Link 2)

V4 DATA

V2 V3 V4

OVERVIEW

Released in April of 2022. Our latest and greatest dataset has 1191 features and targets for all of history.

If you're not sure where to start, take a look at the example-scripts repository.

Otherwise you can download data directly from our API using numerapi:

```
> pip install numerapi
```

```
from numerapi import NumerAPI
napi = NumerAPI()
napi.download_dataset("v4/train.parquet", "train.parquet")
napi.download_dataset("v4/validation.parquet", "validation.parquet")
napi.download_dataset("v4/live.parquet", "live.parquet")
napi.download_dataset("v4/live_example_preds.parquet", "live_example_preds.parquet")
napi.download_dataset("v4/validation_example_preds.parquet", "validation_example_preds.parquet")
napi.download_dataset("v4/features.json", "features.json")
```

TRAIN.PARQUET

Data used to train your model

VALIDATION.PARQUET

Data used to validate or train your model. This expands every week.

Contains targets for eras that in the V3 dataset were marked as test and didn't previously have targets provided. Expands every week to include data for the most recent eras with targets available.

Predictions on this data are not supported by the diagnostics tool yet.

LIVE.PARQUET

The live data that your model predicts on. This changes every week.

LIVE_EXAMPLE_PREDS.PARQUET

An example file with predictions on the live.parquet file. Allows you to easily try your first submission.

VALIDATION_EXAMPLE_PREDS.PARQUET

An example file with predictions on the validation.parquet file. Not fully supported by our diagnostics tool yet.

FEATURES.JSON

JSON file with statistics about each feature in the dataset and some pre-made feature sets to make modeling easier.

Removing Dangerous Features (Link to Forum)

Removing Dangerous Features

Data Science

These 5 are the worst offenders, but we also suggest removing 5 more features as well due to a similar reason. Below are the complete lists of features to remove from your models in the v4 and v3 datasets.

v4:

```
['feature_palpebral_univalve_pennoncel',
 'feature_unsustaining_chewier_adnoun',
 'feature_brainish_nonabsorbent_assurance',
 'feature_coastal_edible_whang',
 'feature_disprovable_topmost_burrower',
 'feature_trisomic_hagiographic_fragrance',
 'feature_queenliest_childing_ritual',
 'feature_censorial_leachier_rickshaw',
 'feature_daylong_ecumenic_lucina',
 'feature_steric_coxcombic_relinquishment']
```

v3:

```
['feature_base_ingrain_calligrapher',
 'feature_unvaried_social_bangkok',
 'feature_deliberative_connatural_kinetoscope',
 'feature_haziest_lifelike_horseback',
 'feature_accusatory_disinfectant_deportment',
 'feature_exorbitant_myeloid_crinkle',
 'feature_jerkwater_eustatic_electrocardiograph',
 'feature_undivorced_unsatisfying_praetorium',
 'feature_direst_interrupted_paloma',
 'feature_lofty_acceptable_challenge']
```

These features are not present in v2 data.

We will not be changing the construction of the v3 and v4 datasets, because we are very sensitive to disrupting user pipelines.

Future data releases will not include any of these features of course.

Baseline Model

Baseline LightGBM with V4 Medium (Safe) Features

- Original V4 medium feature set = 472 features
- Removing 4 dangerous features
- V4 medium (safe) features = **468 features**
- Build a baseline gradient boosting model with same parameters from Numerai example script (rank average from three runs with different seeds):

```
model_params = {"n_estimators": 2000,
                "learning_rate": 0.01,
                "max_depth": 5,
                "num_leaves": 2 ** 5,
                "colsample_bytree": 0.1}
```

Validation: Blocked Time Series Cross-Validation

Using all training and validation data

V4 training + validation up to era 1000
(it is expanding every week)

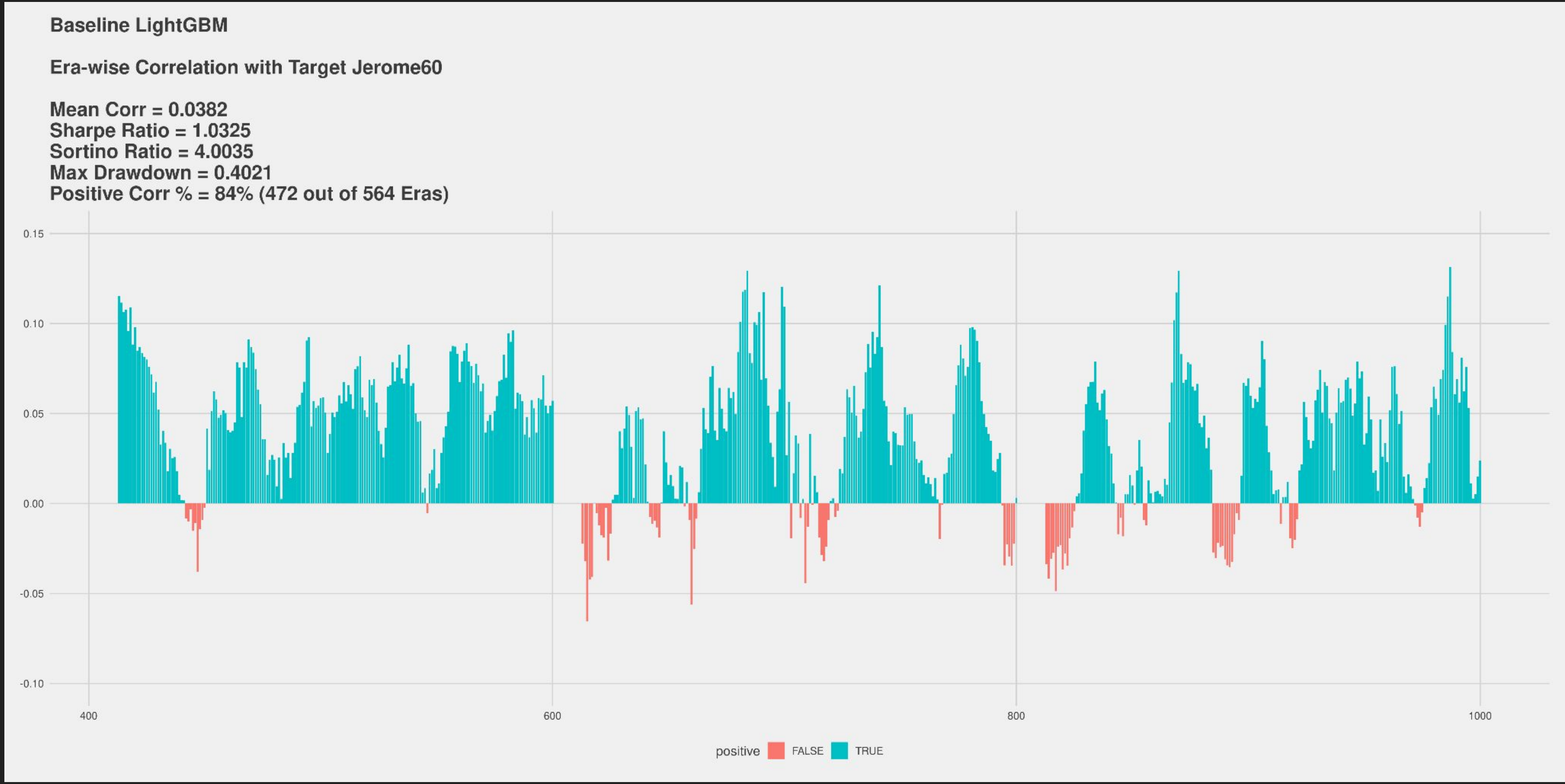
Split into 5 groups with a 12-week (era) gap - avoid data leakage

Group 1	Group 2	Group 3	Group 4	Group 5
Era 1-200	Era 213-400	Era 413-600	Era 613-800	Era 813-1000

Blocked Time Series CV

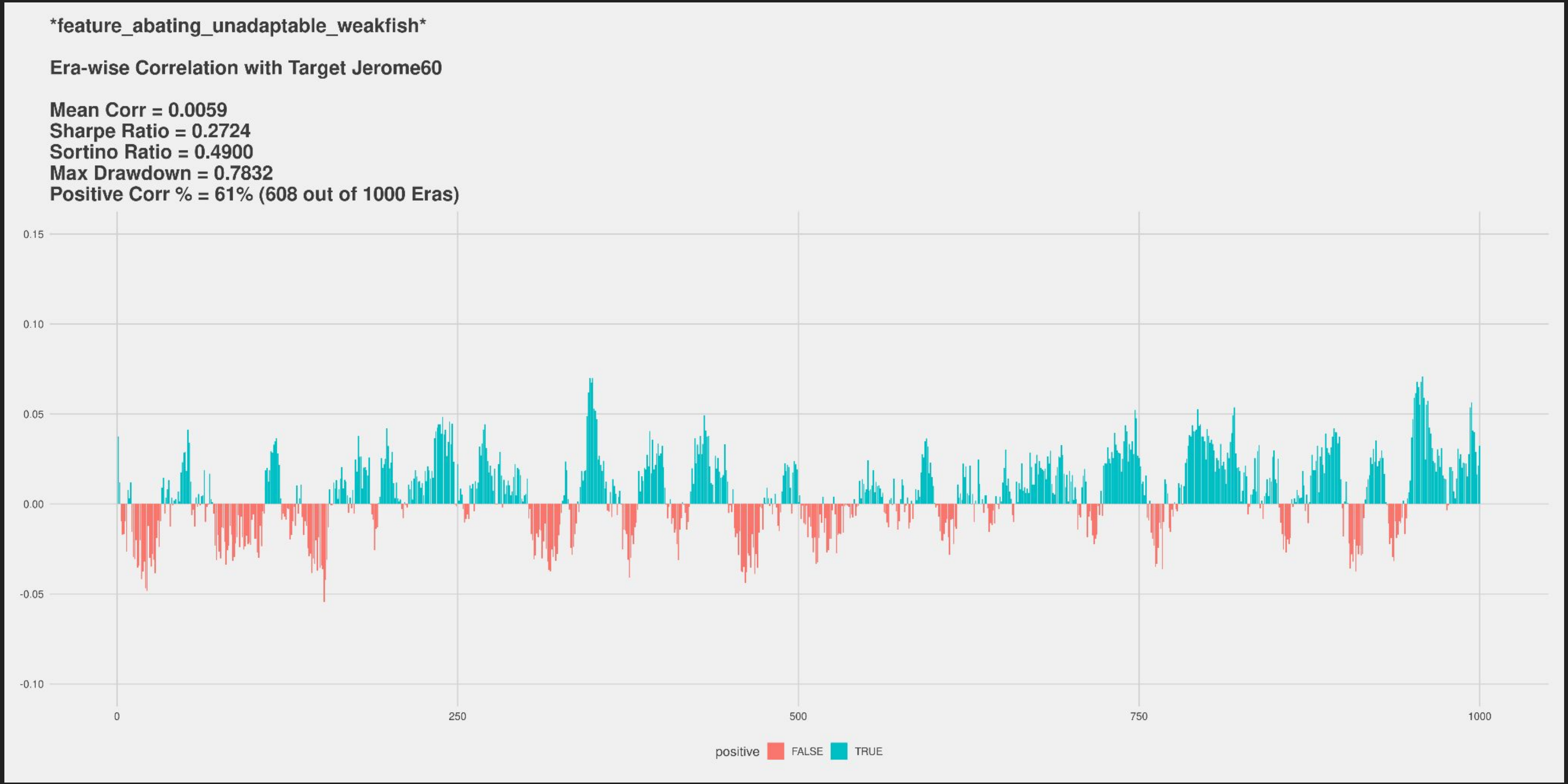
Training Era 1-200	Tuning Era 213-400	Validation Era 413-600		
	Training Era 213-400	Tuning Era 413-600	Validation Era 613-800	
		Training Era 413-600	Tuning Era 613-800	Validation Era 813-1000

Baseline LightGBM with v4 Medium (Safe) Features



Stable & Unstable Features

Evaluating Individual Features

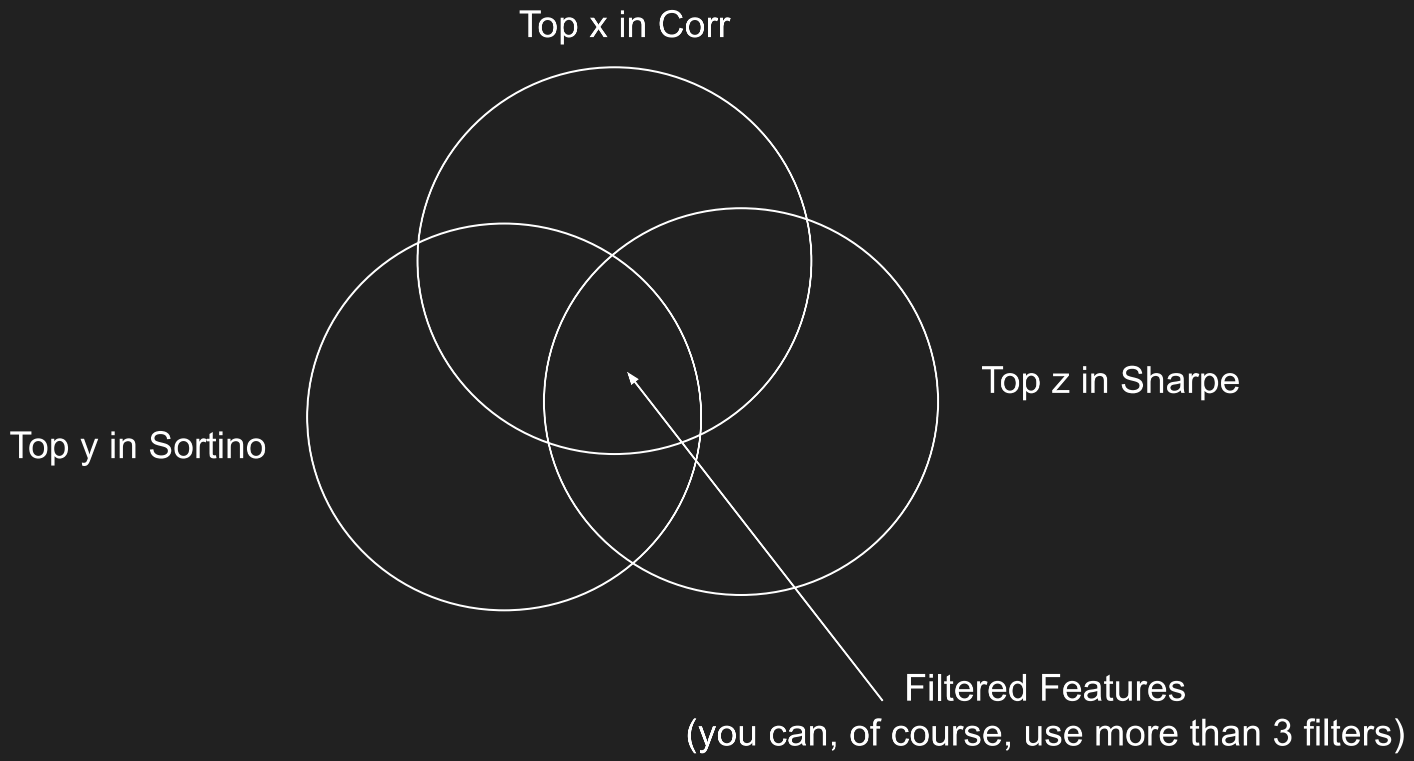


Rank by Corr, Std, Sharpe, Sortino, and Max Drawdown

	feature	cor	sd	sharpe	sortino	maxdd	rank_cor	rank_sd	rank_sharpe	rank_sortino	rank_maxdd
1:	feature_abating_unadaptable_weakfish	0.0058648551	0.02152917	0.27241442	0.49003223	0.7832322	178	165	144	139	235
2:	feature_ablest_mauritanian_elding	0.0162955687	0.02421865	0.67285207	1.96836043	0.2399471	20	284	12	9	2
3:	feature_acclimatisable_unfeigned_maghreb	0.0125839052	0.02646177	0.47555045	1.02767308	0.4622162	43	350	56	57	51
4:	feature_accommodable_crinite_cleft	0.0046452352	0.02200797	0.21107063	0.35214637	0.7531785	227	188	198	200	201
5:	feature_acetose_periotic_coronation	0.0047094414	0.04180363	0.11265628	0.17361047	0.9709245	223	456	321	322	453

468:	feature_witchy_orange_muley	0.0001985314	0.01977576	0.01003913	0.01425890	0.9125060	465	79	462	463	388
469:	feature_wombed_liberatory_malva	0.0008146111	0.05884680	0.01384291	0.01963721	0.9968782	425	467	457	457	469
470:	feature_wrathful_prolix_colotomy	0.0057223103	0.01743858	0.32814077	0.60795344	0.6626324	184	14	105	105	131
471:	feature_wrought_muckier_temporality	0.0017405724	0.02271512	0.07662616	0.11894898	0.9658311	371	220	374	373	447
472:	feature_yelled_hysteretic_eath	0.0005043488	0.02663458	0.01893586	0.02616075	0.9441536	444	354	447	448	423

Feature Selection by Filtering



Optimization

Feature Selection + Parameters Tuning in one go

The Optimization Function

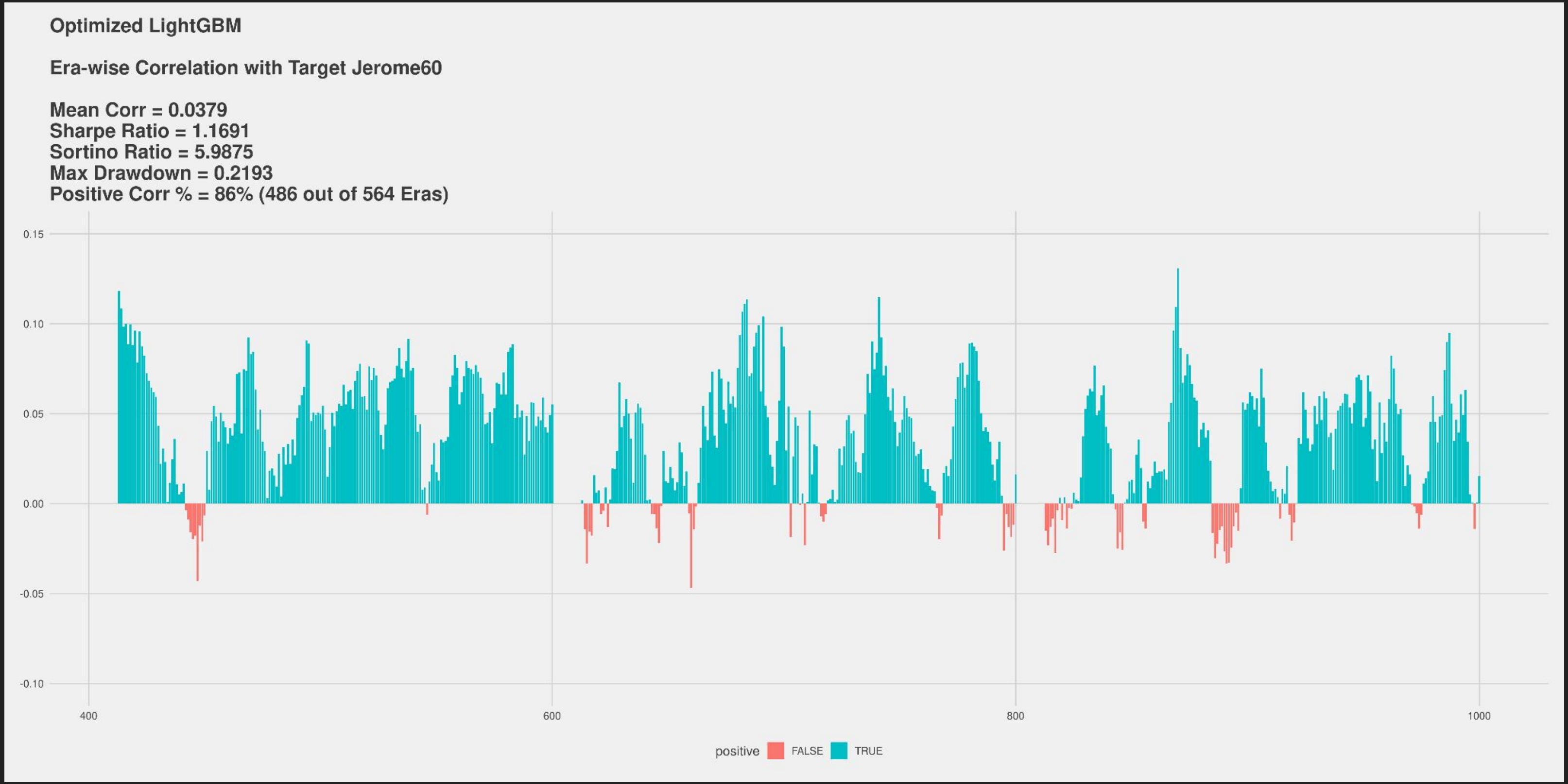
```
scoringFunction = function(# Feature Selection by Filtering
                           top_cor,
                           top_shp,
                           top_srt,

                           # Param tuning for specific algo
                           max_depth,
                           num_leaves,
                           ...
                           bagging_fraction,
                           feature_fraction)
```

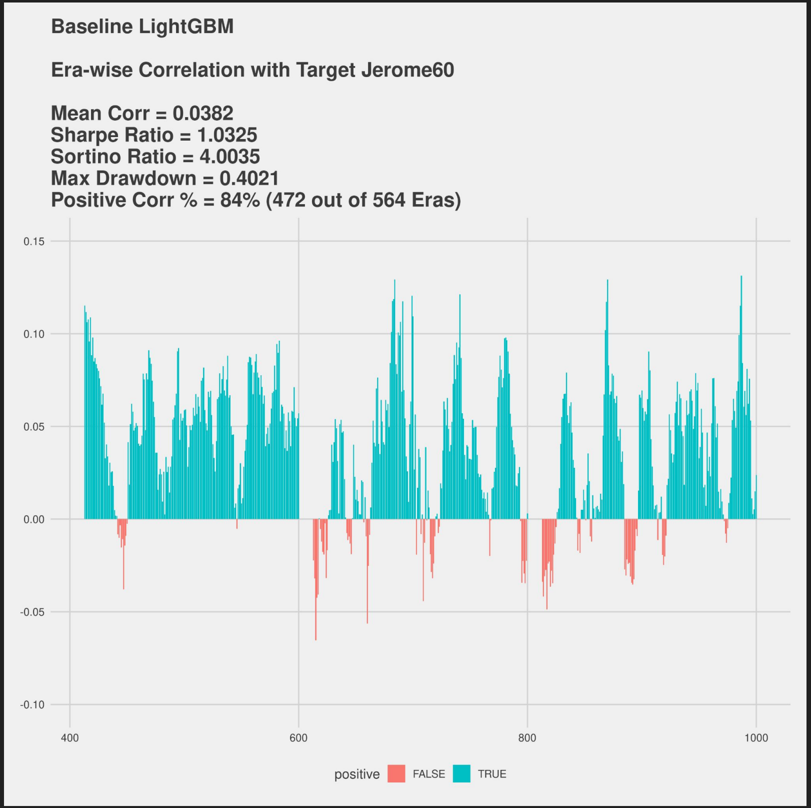
Note:

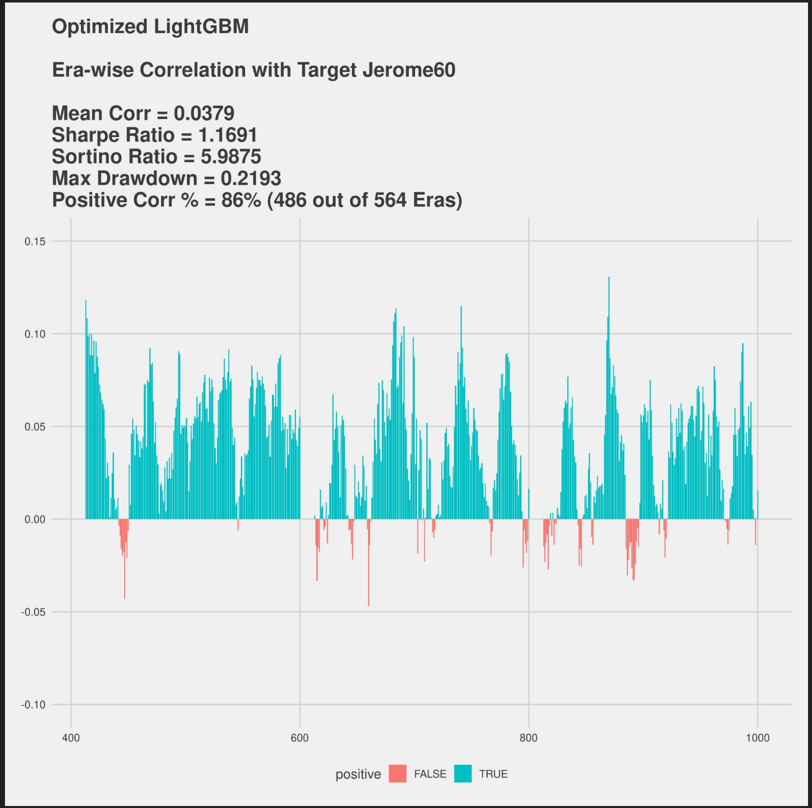
1) It can return any metric I need (e.g. Sharpe, Sortino, FNC, FNC+TB500 …)
2) I use Bayesian Optimization

One of the Results (Optimized for Sharpe Ratio)



Quick Comparison (Baseline vs. Optimized)





My Models (Mostly Optimized for FNC)

Leaderboard THE_

		CORR	CORJ60	FNC	▾ FNCV3	TC
43	THE_COIN_01	0.0413	0.0380	0.0280	0.0285	0.0079
44	THE_COIN_05 2XTC	0.0383	0.0305	0.0283	0.0285	0.0141
63	THE_COIN_02	0.0373	0.0335	0.0248	0.0274	0.0163
68	THE_PUMP_01	0.0344	0.0342	0.0262	0.0272	0.0238
110	THE_COIN_04	0.0257	0.0147	0.0252	0.0267	0.0180
112	THE_COIN_03	0.0281	0.0149	0.0256	0.0267	0.0204
122	THE_FARMER_01	0.0377	0.0423	0.0277	0.0265	0.0133
132	THE_BLENDER_01	0.0371	0.0372	0.0281	0.0263	0.0099
181	THE_PUMP_04	0.0419	0.0516	0.0298	0.0257	0.0056
185	THE_PUMP_05 2XTC	0.0398	0.0381	0.0300	0.0256	0.0268
251	THE_BLENDER_03	0.0339	0.0357	0.0260	0.0248	0.0048
255	THE_FARMER_03	0.0390	0.0489	0.0255	0.0247	0.0005
308	THE_FARMER_04	0.0389	0.0397	0.0274	0.0244	0.0134
316	THE_FARMER_05	0.0392	0.0280	0.0285	0.0243	0.0204

Conclusion

- Simple feature selection by filtering
- Optimizing both feature selection and algo parameters at the same time
- Goal: stable predictions (high Sharpe, high Sortino, low Max Drawdown)

Thanks!

Any Questions?

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